

IBM APPLIED DATA SCIENCE CAPSTONE PROJECT

AYUSH ARJUN PANDE

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GITHUB LINK: <https://github.com/ayush-1105/IBM-APPLIED-DATA-SCIENCE-CAPSTONE>

Executive Summary

Objective:

Predict if SpaceX Falcon 9 first stage will land successfully

Methodologies Used:

- Data Collection – SpaceX API & Web Scraping
- Data Wrangling – Cleaning & Labeling
- EDA – SQL & Data Visualization
- Interactive Analytics – Folium & Plotly Dash
- Machine Learning – Classification Models

Results:

- KSC LC-39A is the best launch site
- Success rate grew from 0% (2010) to 90% (2019)
- Decision Tree gave highest accuracy of 91.4%

INTRODUCTION

Objective:

Analyze SpaceX Falcon 9 launch data to predict first stage landing success

Why It Matters:

- First stage landing = rocket reuse = cost savings
- SpaceX charges \$62M per launch
- If first stage recovered, cost drops to \$28M

Desirable Outcomes:

- Find features that affect landing success
- Track success rate over the years
- Build an accurate ML prediction model

Data collection methodology

Source 1 – SpaceX REST API:

- Requested data from SpaceX API
- Parsed JSON into Pandas DataFrame
- Filtered only Falcon 9 launches
- Replaced missing values with column mean
- And Exported as CSV

Source 2 – Web Scraping (Wikipedia):

- Here we Used BeautifulSoup to parse HTML
- Extracted data from table headers
- Created Pandas Dataframe
- In last Exported as CSV

GITHUB CODE: https://github.com/ayush-1105/IBM-APPLIED-DATA-SCIENCE-CAPSTONE/tree/main/capstone_project

DATA COLLECTION API RESULT

The below shows a few rows of the data collected from the SpaceX API that only contains information pertaining to Falcon 9 Booster Version as shown in the dataframe

FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	
4	1	2010-06-04	Falcon 9	NaN	LEO	CCSFS SLC 40	None None	1	False	False	False
5	2	2012-05-22	Falcon 9	525.0	LEO	CCSFS SLC 40	None None	1	False	False	False
6	3	2013-03-01	Falcon 9	677.0	ISS	CCSFS SLC 40	None None	1	False	False	False
7	4	2013-09-29	Falcon 9	500.0	PO	VAFB SLC 4E	False Ocean	1	False	False	False
8	5	2013-12-03	Falcon 9	3170.0	GTO	CCSFS SLC 40	None None	1	False	False	False

WEB SCRAPPING

The below shows the data that was collected from the Wikipedia to SpaceX and it is shown in the data frame using the pandas

	Flight No.	Launch site	Payload	Payload mass	Orbit	Customer	Launch outcome	Version Booster	Booster landing	Date	Time
0	1	CCAFS	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success\n	F9 v1.0B0003.1	Failure	4 June 2010	18:45
1	2	CCAFS	Dragon	0	LEO	NASA	Success	F9 v1.0B0004.1	Failure	8 December 2010	15:43
2	3	CCAFS	Dragon	525 kg	LEO	NASA	Success	F9 v1.0B0005.1	No attempt\n	22 May 2012	07:44
3	4	CCAFS	SpaceX CRS-1	4,700 kg	LEO	NASA	Success\n	F9 v1.0B0006.1	No attempt	8 October 2012	00:35
4	5	CCAFS	SpaceX CRS-2	4,877 kg	LEO	NASA	Success\n	F9 v1.0B0007.1	No attempt\n	1 March 2013	15:10
...	...	...	...	...	...	...	...	...	...	...	...
237	117	CCSFS	Starlink	15,600 kg	LEO	SpaceX	Success\n	F9 B5B1051.10	Success	9 May 2021	06:42
238	118	KSC	Starlink	~14,000 kg	LEO	SpaceX	Success\n	F9 B5B1058.8	Success	15 May 2021	22:56

DATA WRANGLING MYTHOLOGY

Steps Performed:

- Initial EDA on raw dataset
- We Summarized launches per site
- And also Summarized orbit type occurrences
- Then Summarized mission outcomes

Label Created: concluded the following things

1 = Successful Landing

0 = Unsuccessful Landing

For better understanding of the model

Purpose:

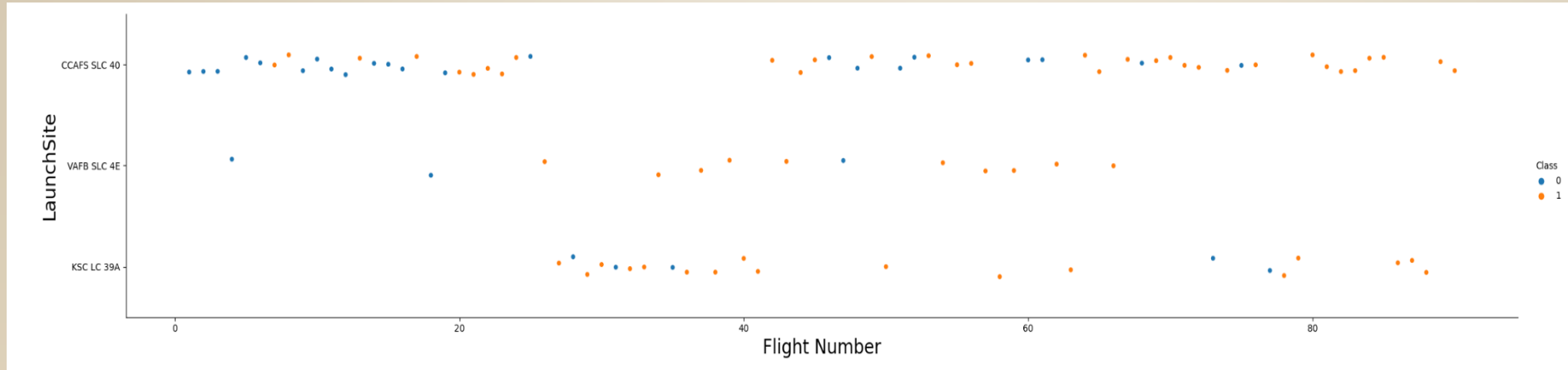
For further operations on data we needed Clean and structured data ready for EDA and ML modeling

GITHUB CODE: https://github.com/ayush-1105/IBM-APPLIED-DATA-SCIENCE-CAPSTONE/tree/main/capstone_project

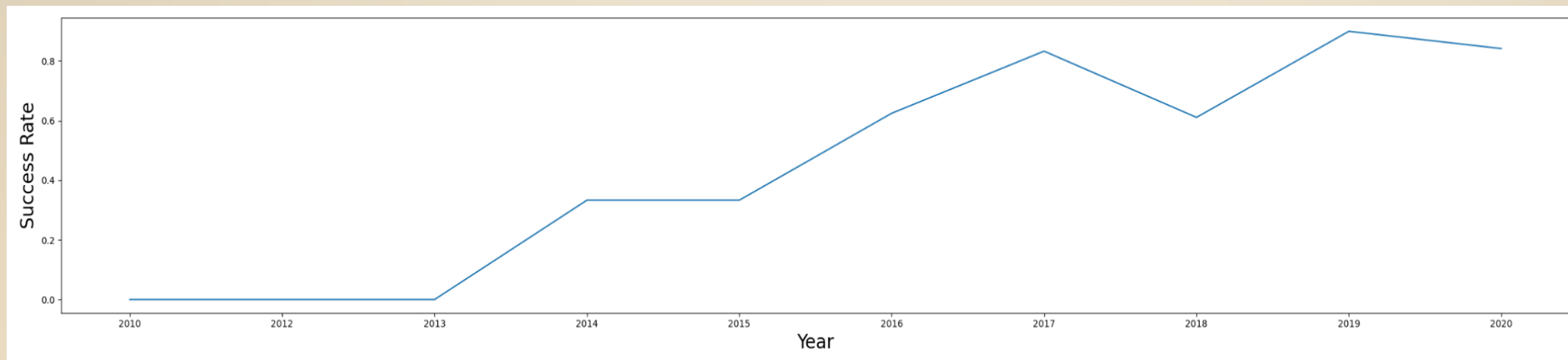


EDA with Data Visualization

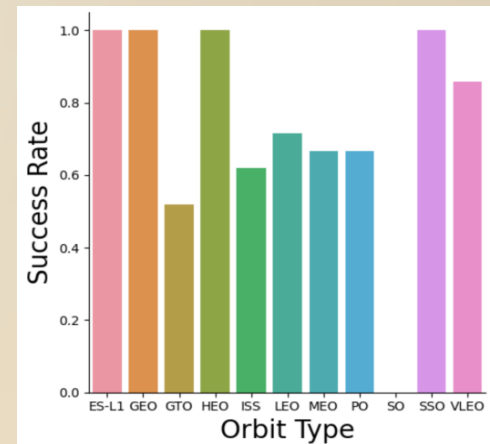
Scatterplots, Bar plots and line plots were plotted to verify and visualize the correlation of features.



3. Line Plot: Success Rate vs Year



Bar Chart Plot: Success Rate vs Orbit Type



INTERACTIVE MAPPING USING THE FOLIUM

Library Integration: Utilized the Folium Python library to generate interactive Leaflet.js map tiles for geospatial data visualization.

Geospatial Feature Mapping:

Point Markers: Plotted specific latitude/longitude coordinates (e.g., launch pad locations) as distinct map markers for visual identification.

Circular Highlights: Implemented circular buffers around launch sites to provide a focus area and highlight specific zones of interest.

Marker Clustering: Applied the MarkerCluster function to group high-density launch events occurring at identical coordinates, preventing visual clutter while maintaining data integrity.

Distance Analysis: Programmatically calculated the Euclidean distance between launch pad coordinates and the coastline, visualized using integrated PolyLine overlays to validate safety constraints.



Plotly Dash for dashboard

Dashboard Framework: Developed an interactive dashboard interface using the Plotly Dash library to aggregate and visualize mission performance data.

Core Visualization Components:

Dynamic Pie Charts: Generated proportional success metrics filtered by specific launch sites via an integrated dropdown menu.

Interactive Payload Correlation: Implemented a scatter plot to analyze the relationship between PayloadMass and Class (Landing Success).

User-Defined Inputs: Enabled an interactive RangeSlider to allow for granular filtering of payload mass (kg), providing real-time data exploration.

Analytical Objective:

The dashboard serves to isolate the correlation between payload mass and landing reliability, enabling data-driven identification of the optimal launch facility to maximize the probability of first-stage recovery.



PREDICTIVE ANALYSIS (CLASSIFICATION)

Model Selection: Employed four distinct supervised learning algorithms to classify landing outcomes (0 = Unsuccessful, 1 = Successful):

Logistic Regression: Used as a baseline for binary classification performance.

Support Vector Machines (SVM): Implemented to find the optimal hyperplane for separating success/failure classes in high-dimensional feature spaces.

Decision Tree: Utilized for non-linear classification to capture complex feature interactions within the launch parameters.

K-Nearest Neighbours (KNN): Applied to classify outcomes based on the proximity of mission data to historical feature vectors.



EXPLORATORY DATA ANALYSIS(EDA) USING SQL

- Extracted distinct launch facility names utilized across all space missions.
- Filtered and retrieved the first five mission records originating from launch sites with the 'CCA' prefix.
- Calculated the aggregate payload mass transported by boosters specifically for NASA (CRS) missions.
- Computed the average payload mass capacity specifically for the F9 v1.1 booster configuration.
- Identified the precise chronological date of the inaugural successful ground pad landing.
- Isolated successful booster versions carrying payload masses strictly within the 4,000 to 6,000 kg threshold.
- Aggregated and quantified the total volume of successful versus failed mission outcomes.
- Utilized nested subqueries to dynamically extract the booster versions responsible for transporting the maximum recorded payload mass.
- Extracted monthly records detailing drone ship landing failures, including corresponding booster versions and launch sites, specifically for the year 2015.
- Ranked the frequency of successful landing outcomes in descending order for the temporal window between June 4, 2010, and March 20, 2017.

EXPLORATORY DATA ANALYSIS(EDA) USING SQL

- ❖ Calculated the aggregate payload mass transported by boosters for NASA (CRS) missions.

```
%sql select sum(payload_mass__kg_) as total_payload_mass from SPACEX where customer = 'NASA (CRS)';
```

```
* ibm_db_sa://wfd98799:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8lcg.databases.appdomain.cloud:31198/bludb  
Done.
```

total_payload_mass
45596

- ❖ Computed the average payload mass capacity specifically for the F9 v1.1 booster configuration.

```
In [12]: %sql select avg(payload_mass__kg_) as average_payload_mass from SPACEX where booster_version like '%F9 v1.1%';
```

```
* ibm_db_sa://wfd98799:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8lcg.databases.appdomain.cloud:31198/bludb  
Done.
```

```
Out[12]: average_payload_mass
```

2534

EXPLORATORY DATA ANALYSIS(EDA) USING SQL

Filtered and retrieved launch site names beginning with the 'CCA' prefix.

```
%sql select * from SPACEX where launch_site like 'CCA%' limit 5;
```

* ibm_db_sa://wfd98799:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8l1cg.databases.appdomain.cloud:31198/bludb
Done.

DATE	time_utc	booster_version	launch_site	payload	payload_mass_kg	orbit	customer	mission_outcome	landing_outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

❖ Extracted all distinct launch site names via SQL querying.

```
In [9]: %sql select distinct launch_site from SPACEX;
```

* ibm_db_sa://wfd98799:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8l1cg.databases.appdomain.cloud:31198/bludb
Done.

```
Out[9]: launch_site
```

CCAFS LC-40
CCAFS SLC-40
KSC LC-39A
VAFB SLC-4E

EXPLORATORY DATA ANALYSIS(EDA) USING SQL

- ❖ Identified the exact chronological date of the inaugural successful ground pad landing.

```
In [13]: %sql select min(date) as first_successful_landing from SPACEX where landing__outcome = 'Success (ground pad)';

* ibm_db_sa://wfd98799:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8lcg.databases.appdomain.cloud:31198/bludb
Done.

Out[13]: first_successful_landing
          2015-12-22
```

Isolated booster versions that achieved drone ship landing success while carrying payloads between 4,000 and 6,000 kg.

```
In [14]: %sql select booster_version from SPACEX where landing__outcome = 'Success (drone ship)' and payload_mass__kg_ between 4000 and 6000;

* ibm_db_sa://wfd98799:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8lcg.databases.appdomain.cloud:31198/bludb
Done.

Out[14]: booster_version
          F9 FT B1022
          F9 FT B1026
          F9 FT B1021.2
          F9 FT B1031.2
```


EXPLORATORY DATA ANALYSIS(EDA) USING SQL

- ❖ Retrieved detailed records—including month names, drone ship landing failures, booster versions, and launch sites—specifically for the 2015 calendar year.

```
%%sql select monthname(date) as month, date, booster_version, launch_site, landing__outcome from SPACEX
where landing__outcome = 'Failure (drone ship)' and year(date)=2015;
```

```
* ibm_db_sa://wfd98799:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90108kqb1od8lcg.databases.appdomain.cloud:31198/blddb
Done.
```

MONTH	DATE	booster_version	launch_site	landing_outcome
January	2015-01-10	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
April	2015-04-14	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

- ❖ Ranked the frequency of successful landing outcomes in descending order for the period between June 4, 2010, and March 20, 2017.

```
%%sql select landing__outcome, count(*) as count_outcomes from SPACEX
where date between '2010-06-04' and '2017-03-20'
group by landing__outcome
order by count_outcomes desc;
```

```
* ibm_db_sa://wfd98799:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90108kqb1od8lcg.databases.appdomain.cloud:31198/blddb
Done.
```

landing_outcome	count_outcomes
No attempt	10
Failure (drone ship)	5
Success (drone ship)	5
Controlled (ocean)	3
Success (ground pad)	3
Failure (parachute)	2
Uncontrolled (ocean)	2
Precluded (drone ship)	1

EXPLORATORY DATA ANALYSIS(EDA) USING SQL

- ❖ Aggregated and quantified the total count of successful versus failed mission outcomes.

```
%sql select mission_outcome, count(*) as total_number from SPACEX group by mission_outcome;
```

```
* ibm_db_sa://wfd98799:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8l1cg.databases.appdomain.cloud:31198/bludb  
Done.
```

mission_outcome	total_number
Failure (in flight)	1
Success	99
Success (payload status unclear)	1

- ❖ Utilized subqueries to dynamically retrieve the booster versions that have transported the maximum recorded payload mass.

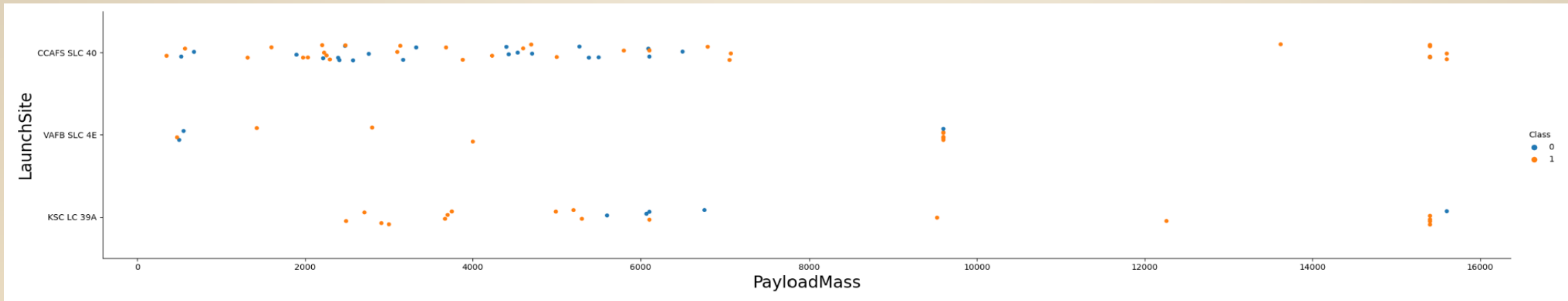
```
%sql select booster_version from SPACEX where payload_mass_kg_ = (select max(payload_mass_kg_) from SPACEX);
```

```
* ibm_db_sa://wfd98799:***@0c77d6f2-5da9-48a9-81f8-86b520b87518.bs2io90l08kqb1od8l1cg.databases.appdomain.cloud:31198/bludb  
Done.
```

booster_version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

EDA with Visualization

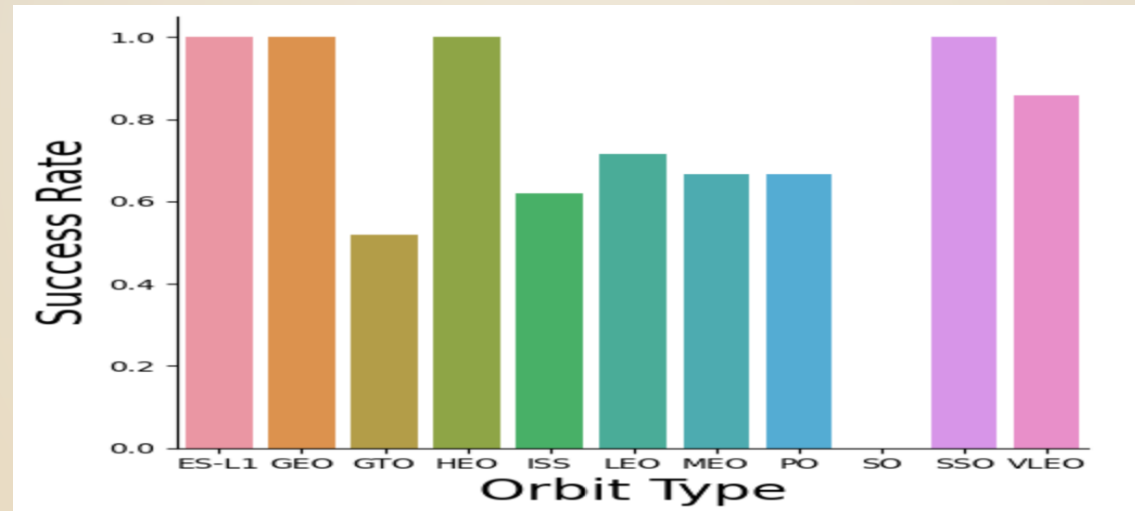
- Analyzes payload capacity against landing success (\$I\$) or failure (\$0\$). "KSC LC 39A" demonstrates a superior success rate; meanwhile, "CCAFS SLC 40" is primarily utilized for lower-mass payloads, which correlate with lower recovery success



EDA with Visualization

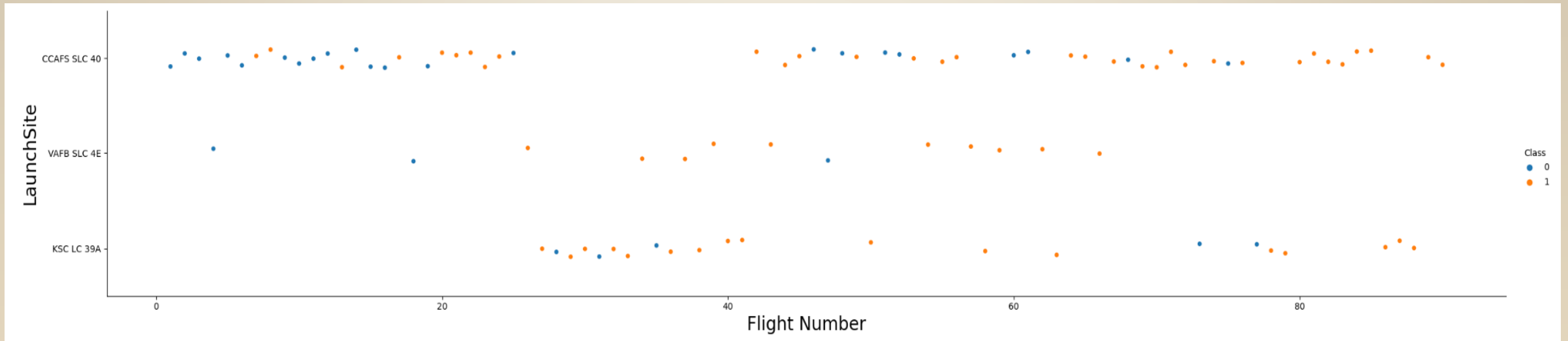
Success Rate vs. Orbit Type

- Evaluates recovery performance across various orbital trajectories. "ES-L1," "GEO," "HEO," and "SSO" achieve a 100% success rate, while "VLEO" maintains 85% and "SO" currently records 0%.



EDA with Visualization

Launch Site vs. Flight Number

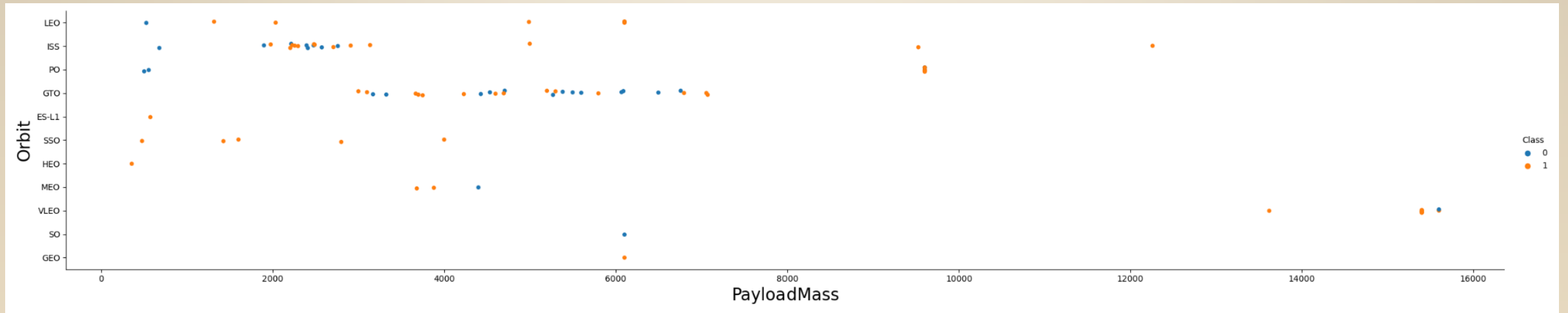


- Maps landing outcomes (\$0=\$ Unsuccessful, \$1=\$ Successful) across the flight history.“
- CCAFS SLC 40" exhibits the highest volume of recent successful recoveries, followed by "KSC LC 39A" and "VAFB SLC 4E".

EDA with Visualization

Orbit vs. Payload Mass

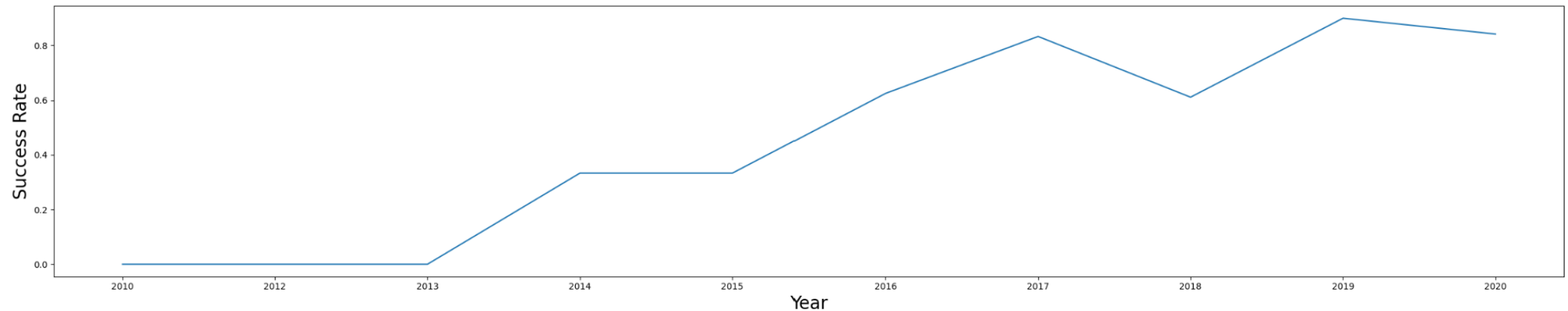
- Correlates orbital targets with payload weight capacity and recovery status. Payloads under 8,000 kg are predominantly directed to "LEO," "ISS," and "GTO," whereas heavier configurations are increasingly targeted for "VLEO," demonstrating a higher success rate for high-mass vehicles.



EDA with Visualization

Success Rate vs. Year

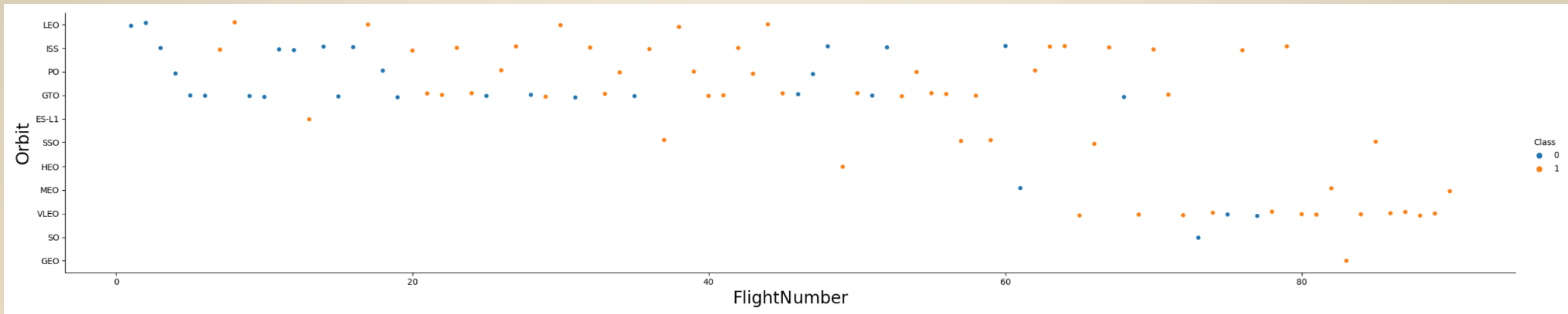
- Tracks the long-term evolution of recovery reliability from 2010 to 2020.
- The program achieved 90% success in 2019, with the most critical maturation period occurring between 2013 and 2014, signaling consistent progress toward total recovery reliability.



EDA with Visualization

Orbit vs. Flight Number

- Tracks landing outcomes relative to mission targets across the flight timeline. Data reveals a strategic pivot toward "VLEO" and "MEO" orbits in recent missions, correlating with a notable increase in recovery reliability.



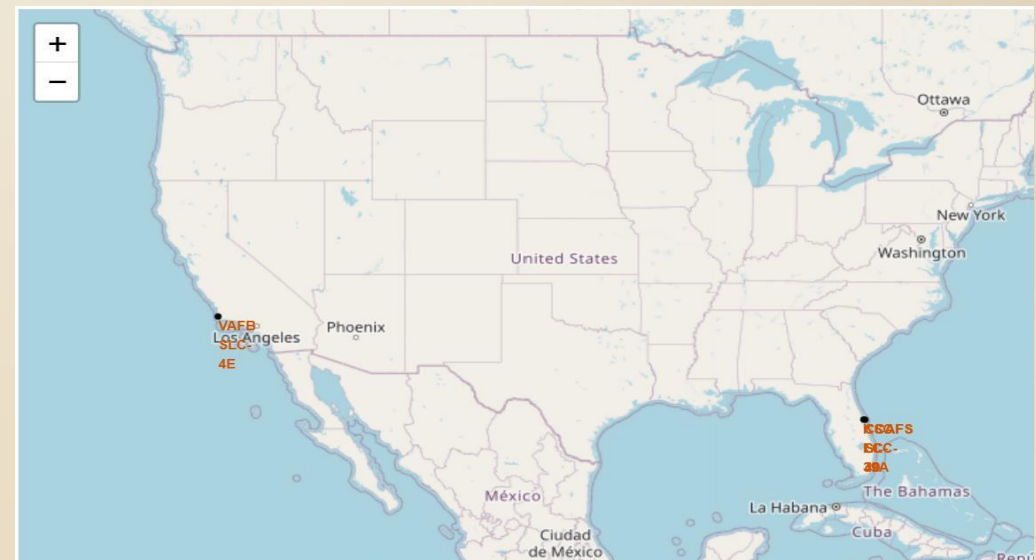
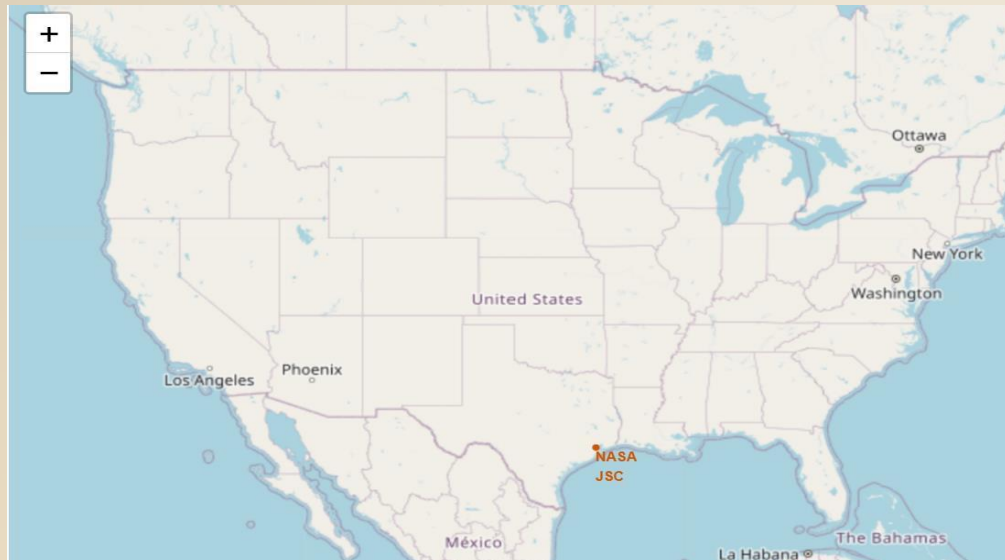
Interactive map results using Folium

FOLLOWING LAUNCH SITES

Coastal Positioning: Facilities are placed along coastlines to ensure flight paths remain over open water, mitigating the risk of debris impacting populated land.

Geographical Spread: Sites are strategically distributed across both the East and West coasts to optimize trajectory accessibility for diverse orbital missions.

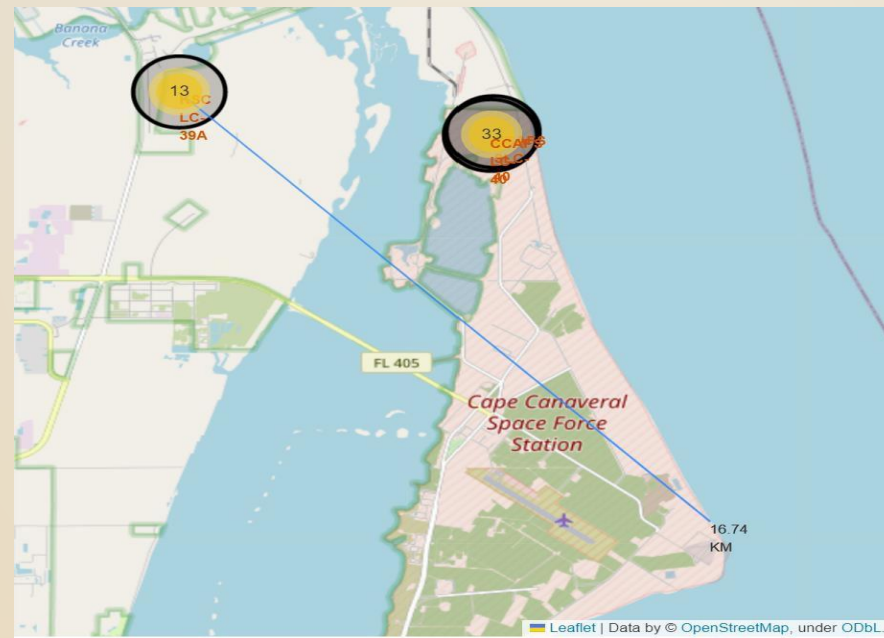
Interactive Mapping: Using Folium, the map allows for seamless zooming and panning to inspect specific launch coordinates with high precision.



Interactive map results using Folium

Safety Buffer: The distance between the KSC LC-39A launch site and the Cape Canaveral coastline is explicitly measured to ensure safe flight trajectories.

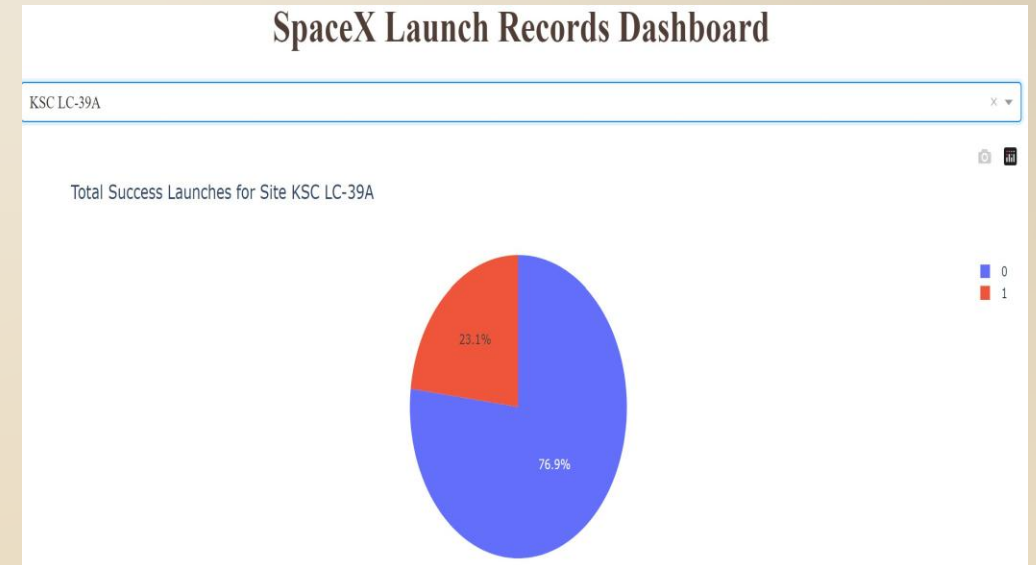
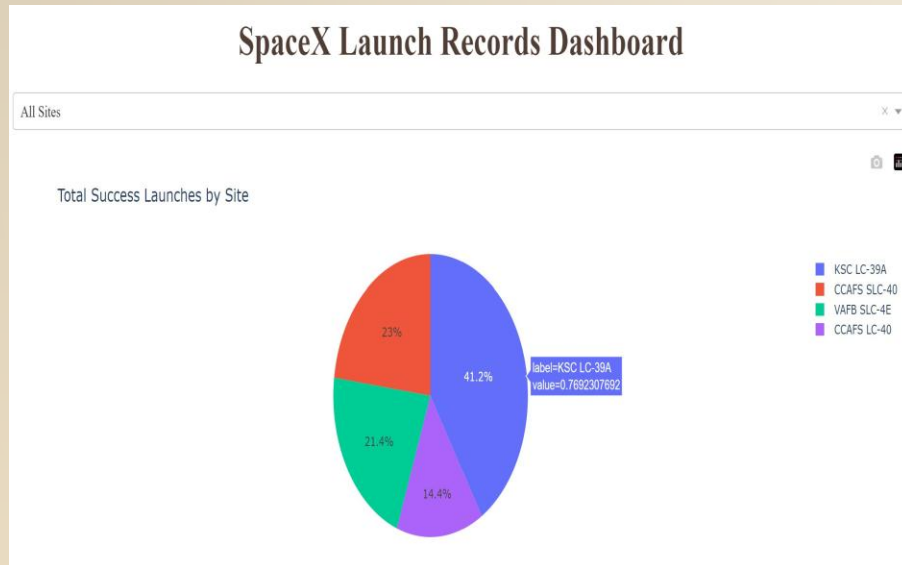
Precise Geometry: A 16.74 km separation line is mapped to visually validate that the launch pad remains at a safe, calculated distance from the coast.



Plotly Dash dashboard results

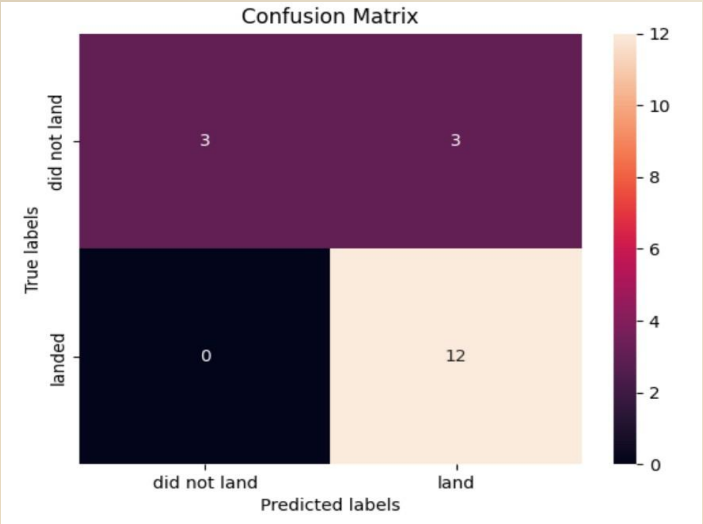
Global Success Distribution: The dashboard analysis confirms KSC LC-39A as the primary launch site, contributing 41.2% to the total volume of successful mission recoveries.

Site-Specific Reliability: When isolating the KSC LC-39A facility, operational data reveals a high-performance reliability metric of 76.9% successful landings compared to 23.1% failures.



Predictive analysis (classification) results

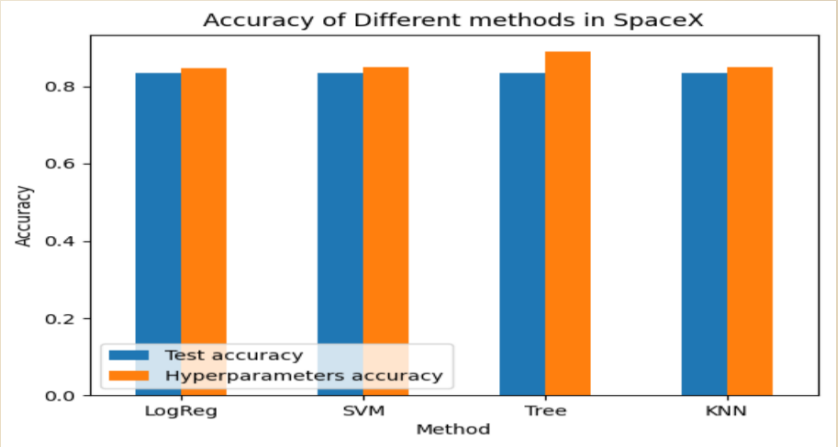
This matrix provides a granular assessment of model performance by cross-referencing true landing outcomes against predicted labels.



Confusion Matrix

The Decision Tree classifier demonstrated superior predictive reliability, achieving the highest test and hyperparameter-tuned accuracy scores among all evaluated models.

	LogReg	SVM	Tree	KNN
Test Accuracy	0.833333	0.833333	0.888889	0.833333
Hyperparametric accuracy	0.846429	0.848214	0.914286	0.848214



Accuracy of Data Model

CONCLUSION

1. Orbit Performance: First-stage landings have achieved perfect success rates for trajectories including ES-LI, GEO, HEO, SSO, and VLEO.
2. Operational Trends: Success rates have climbed steadily over the years, peaking at an impressive 98% in 2019.
3. Site Efficiency: The KSC LC-39A facility stands out as the most successful launch site, consistently supported by its strategic coastal location designed to maximize safety.
4. Payload Optimization: Missions carrying payloads between 2,000 and 5,000 kg have proven to be the most reliable configurations for successful landing outcomes.
5. Predictive Modeling: Advanced machine learning analysis confirms high overall accuracy, with the Decision Tree model delivering the most reliable results for predicting future mission success.

THANK YOU



GITHUB CODE: https://github.com/ayush-1105/IBM-APPLIED-DATA-SCIENCE-CAPSTONE/tree/main/capstone_project